

ADITYA ALOK

AI/ML Engineer

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Pune, Maharashtra, India

PROFESSIONAL SUMMARY

Staff AI/ML Engineer with 6+ years of progressive experience architecting, deploying, and scaling enterprise AI solutions across distributed systems. Deep expertise in Large Language Models (LLMs), transformer architecture, Generative AI (including RLHF), RAG pipelines, NLP, and Computer Vision. Proven track record of leading cross-functional teams, driving A/B testing frameworks, and delivering measurable business impact — including \$3M in annual savings and 40% workflow efficiency gains. Experienced across the full MLOps lifecycle on AWS, Azure, and GCP.

TECHNICAL SKILLS

ML & AI	LLM Fine-Tuning, Generative AI, RLHF, Prompt Engineering, RAG, NLP, Transformer Architecture, Computer Vision, Anomaly Detection, Recommendation Systems, A/B Testing
Frameworks	Python, PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, LangChain, Pinecone (Vector DB), LSTM
MLOps	MLflow, Airflow, GitHub Actions, CI/CD, Model Monitoring, Distributed Systems
Cloud	AWS, Azure ML, GCP AI Platform, Terraform
Data Eng.	SQL, Apache Spark, PySpark, Pandas, Snowflake, Data Lakes, ETL Pipelines, Feature Engineering
DevOps	Jenkins, Git, REST APIs, Agile/Scrum, Jira, Docker, Kubernetes

PROFESSIONAL EXPERIENCE

Staff Engineer — AI/ML | Nagarro India

July 2025 – Present

- Lead a team of 5 engineers on enterprise AI initiatives including LLM fine-tuning (transformer-based models) and deployment for automation workflows, improving operational efficiency by 40%.
- Architected end-to-end scalable ML pipelines across distributed systems using MLflow, Docker, and Airflow, reducing model time-to-deploy by 60%.
- Designed an AI-based anomaly detection system with rigorous A/B testing validation, delivering \$3M in annual savings through proactive risk mitigation.
- Established RLHF-aligned evaluation frameworks for LLM outputs, improving response quality and safety for enterprise-facing generative AI applications.
- Mentor 8+ junior and mid-level engineers on ML best practices, transformer architecture, and prompt engineering across squads.

Associate Staff Engineer — AI/ML | Nagarro India

Jan 2024 – July 2025

- Built an NLP-driven document classification system using transformer-based embeddings, improving accuracy by 30% and reducing manual processing effort by 50%.
- Optimized cloud AI infrastructure on AWS SageMaker across distributed training jobs, achieving 20% cost reduction through rightsizing and pipeline efficiency.
- Introduced automated ML testing and A/B testing frameworks to ensure model reliability, reproducibility, and performance in production environments.
- Collaborated with product managers and business stakeholders to align AI roadmaps with enterprise priorities, serving 20,000+ internal users.

Senior Engineer — AI/ML | Nagarro India

July 2022 – Jan 2024

- Designed and deployed a computer vision pipeline for manufacturing defect detection using distributed inference, achieving 95% product quality accuracy across 10M+ items.

- Built a real-time recommendation engine leveraging vector database (Pinecone) similarity search, increasing customer engagement by 25% across 500K+ users.
- Improved ML model performance through advanced feature engineering, hyperparameter tuning, and A/B testing, achieving 15% accuracy improvement across 3 production models.
- Led cross-team integration of AI components into enterprise applications, establishing distributed systems design patterns adopted org-wide.

Engineer — AI/ML | Nagarro India

Feb 2021 – July 2022

- Developed LSTM-based and transformer-assisted predictive models for demand forecasting, reducing supply chain costs by 12% across 50+ SKUs.
- Deployed and maintained scalable ML inference APIs on Kubernetes across distributed clusters, maintaining 99.9% production uptime SLA.
- Partnered with cross-functional teams to integrate AI solutions into core enterprise applications, enabling real-time data pipelines for 3 business units.

Associate Engineer — AI/ML | Nagarro India

June 2019 – Feb 2021

- Contributed to 4 ML prototypes and proof-of-concepts for enterprise clients, reducing manual data processing time by 35% using Python automation.
- Automated data preprocessing pipelines using Python and Apache Spark across distributed data lakes, cutting ETL runtime by 40%.
- Gained hands-on expertise in TensorFlow, PyTorch, transformer models, and Scikit-learn, completing 3 production deployments within first year.

KEY PROJECTS

RAG-Powered Enterprise Chatbot — Built retrieval-augmented generation (RAG) chatbot using LangChain, Pinecone (vector database), and transformer-based embeddings for enterprise knowledge management. Achieved 92% answer relevance in A/B testing against baseline keyword search.

AI-Powered Time Series Forecasting — Delivered LSTM + transformer hybrid solution for demand forecasting across distributed data pipelines, achieving 20% accuracy improvement and \$1.2M projected savings.

MLOps Pipeline Automation — Automated full ML CI/CD lifecycle using GitHub Actions, MLflow, and Kubernetes on distributed cloud infrastructure, enabling continuous model training, versioning, A/B evaluation, and zero-downtime deployment.

EDUCATION

Bachelor of Engineering — Computer Science

2016 – 2019

Dr. APJ Abdul Kalam Technical University, Lucknow | GPA: 7.3/10

CERTIFICATIONS

- Machine Learning Specialization — Stanford University (Andrew Ng)
- DeepLearning.AI Specialization — Coursera
- Generative AI — Google
- SnowPro Core Certification — Snowflake
- Apache Spark and Python for Big Data with Pyspark — Udemy
- DevOps on AWS & Containers on AWS — AWS Training
- Databases and SQL for Data Science — IBM / Coursera

AWARDS & RECOGNITION

- A-Team Award, Nagarro — Recognized 3x for exceptional performance and measurable team contributions.
- Cheer Board Nominations, Nagarro — Acknowledged 6x for innovative AI ideas and outstanding individual impact.
- NAGP (Nagarro Accelerated Growth Program) — Designated top performer demonstrating exceptional leadership and business impact.